



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of CSE (Artificial Intelligence and Machine Learning)

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: I Semester – B.E. CSE (AIML)

Course Code & Title: CS25C01 Computer Programming: C

Name of the Faculty member (s): Dr. R. Venkatesh, ASP & Head /CSE (AIML)

Date of Implementation : 06.10.2025

Innovative Practice Description

Unit / Topic: Module III / Function Parameters and Return Types

Course Outcome: CO 3

Activity Chosen: Role Play

Objective:

The objective of this role-play activity was to help students understand how function parameters, argument passing, and return types work in C programming. Through dramatized role-play, students could visualize how data flows into a function, how processing happens, and how the function returns a value back to the calling function.

Time Allotted for the Activity: 15 minutes

Roles Assigned:

Seven students were given specific roles to represent various components of a C program:

1. Sharmila Devi G – Function Caller (Main Function)
2. Neyathi S – Function Activator (Performs the “call” to the function)
3. Ruthra Devi P – Function Itself (Performs operations)
4. Subhashini S – Parameter 1 (e.g., int a)
5. Gayathri K – Parameter 2 (e.g., int b)
6. Nathira H – Processing Unit (Executes the function logic)
7. Gopika V – Return Value (Carries the output back to main)

Details of the Implementation:

1. Introduction to Problem:

The instructor introduced:

- Types of function parameters:
 - Call by Value
 - Call by Reference
- Return types in C
- How data flows from main() to a function and back
- A simple demonstration function (e.g., `int add(int x, int y)`)
- The concept was explained using a flow diagram, after which the role-play was initiated.

2. Role Assignment:

Students were assigned symbolic roles mimicking how a function works in a C program:

- **Main Function (Sharmila Devi G)** initiates the task by sending inputs.
- **Function Activator (Neyathi S)** simulates the calling mechanism.
- **Parameters (Subhashini S & Gayathri K)** represent the arguments passed.
- **Function Entity (Ruthra Devi P)** receives parameters.
- **Processing Unit (Nathira H)** performs computation.
- **Return Value (Gopika V)** brings back the final output.

3. Activity Execution:

- Step 1: Main Function (Sharmila Devi G) chooses two numbers to send as parameters.
- Step 2: Function Activator (Neyathi S) symbolically “calls” the function.
- Step 3: Parameters (Subhashini & Gayathri) carry the values to the function.
- Step 4: Function (Ruthra Devi P) acknowledges receiving the inputs.
- Step 5: Processing Unit (Nathira H) performs the operation:
 - Example: addition, multiplication, max/min, etc.
- Step 6: Return Value (Gopika V) carries the computed result back to the Main function.
- Step 7: Main Function displays how the returned value is printed using printf.

Throughout the activity, the class observed the movement of “data” and the interaction between students, illustrating the internal working of function call mechanics in C.

4. Completion:

The entire role-play was completed smoothly within the allotted time. Students clearly demonstrated:

- Parameter passing
- Function execution
- Return value mechanism

The concept became visually clear and interactive for the entire class.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PSO1
CO 1	3	3	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5	PSO1
	3	3	3	2	2	2
Justification for correlation	Students applied core programming fundamentals (function calls, argument passing, return mechanisms) to understand modular program design, demonstrating strong conceptual knowledge.	Students analyzed programming problems, identified required parameters, selected appropriate return types, and mapped input-output behavior, showcasing high analytical skill.	Students designed modular solutions using functions, structured code flow, and demonstrated how functions improve clarity and reusability, reflecting strong design capabilities.	Students explored different function behaviors—such as value return, void functions, and multiple parameters—requiring moderate investigative thinking.	Students used C programming tools like IDEs/compiler to implement functions and debug errors, indicating moderate proficiency with programming tools.	Students applied function concepts to write modular C programs, improving real-world problem-solving abilities in computing.

• **Images / Screenshot of the practice:**



Figure 1: Role Play Activity

Learning Outcome:

- **Clear Understanding of Function Mechanism:**
Students understood how parameters are passed and how return values are generated.
- **Improved Programming Logic:**
The visualization helped students write C programs with functions more confidently.
- **Engagement and Participation:**
The dramatized flow encouraged active learning and strengthened conceptual clarity.

Reflective Critique:

Student Feedback:

- Students expressed that the role-play helped them clearly visualize how functions execute in C.
- Many felt it made writing programs with multiple parameters easier.
- They appreciated the clarity in understanding return value behaviour.

Classroom Dynamics:

- The activity promoted teamwork, discussion, and curiosity.
- Students asked more questions and participated actively in the demonstration.

Benefits of the Practice:

- ✓ Improved conceptual clarity
- ✓ Higher engagement through physical demonstration
- ✓ Better understanding of modular programming
- ✓ Encouraged collaborative learning

Challenges Faced:**Limited Time**

- Explaining each role and performing the activity within 15 minutes required careful coordination.

Role Complexity

- Some students required additional guidance to understand the specific role they represented (e.g., Processing Unit, Return Value).

Conclusion:

The role-play activity on **Function Parameters and Return Types in C** was highly effective in enriching student understanding. It transformed an abstract programming concept into a clear and memorable physical demonstration. With seven students actively participating and the entire class observing, the session achieved its goal of increasing engagement, strengthening problem-solving skills, and reinforcing modular programming concepts. Future sessions can involve different function types (void, pointer-returning functions, recursion, etc.) for broader exposure.

References:

- <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
- https://www.ritrjpm.ac.in/images/computer-science/10_CS8591_Rolepaly.pdf
- <https://www.niu.edu/citl/resources/guides/instructional-guide/role-playing.shtml>



Signature of Faculty Member



HOD-CSE (AIML)